

DATA ANALYTICS CASE STUDY

Telco Customer Churn Analysis

Exploratory Data Analysis, Attrition Driver Modeling & Executive Action Plan

 Targeted Strategy to Preserve Recurring Revenue | Portfolio Presentation

Business Understanding

⚠ Problem Statement

Customer churn poses an immediate threat to subscription growth and recurring revenue in telecommunications. Acquiring a new customer costs 5x more than retaining an existing one.

🎯 Business Objective

Identify primary behavioral, demographic, and billing drivers behind churn across 7,043 subscriber accounts to deliver actionable, high-ROI retention interventions.



Dataset Overview & Architecture



Demographics

4 Attributes: Gender, SeniorCitizen status, Partner status, and Dependents coverage.



Services Opted

9 Attributes: Phone, Multiple Lines, Internet (DSL/Fiber Optic), Online Security, Tech Support, Streaming.



Account & Billing

7 Attributes: Contract length, Paperless Billing, Payment Method, Monthly Charges, Total Charges, Tenure.



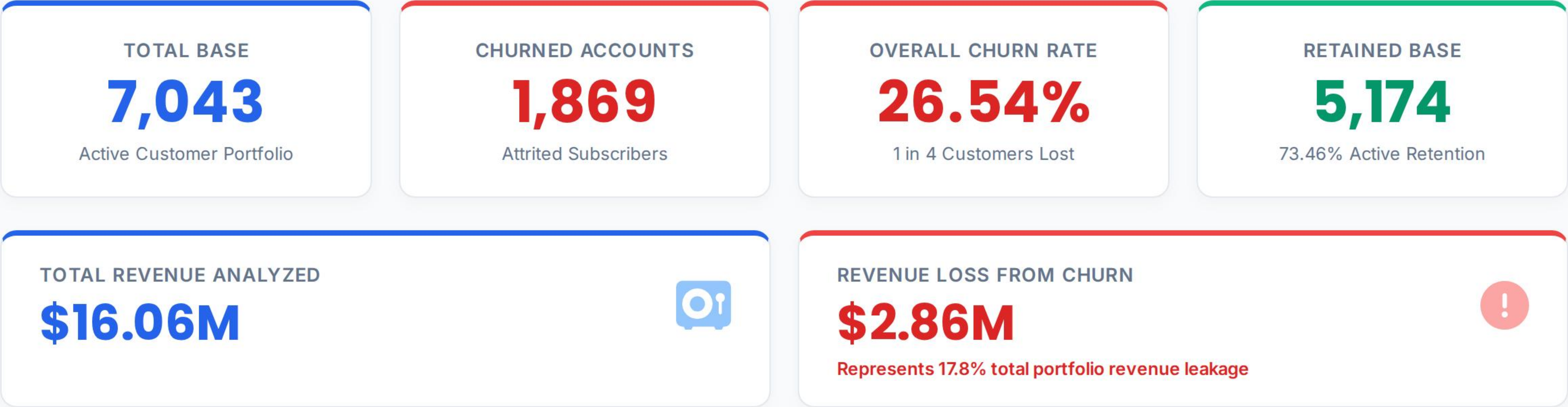
Target Variable

1 Attribute: Binary Churn Flag indicating whether the customer left within the last month (Yes / No).

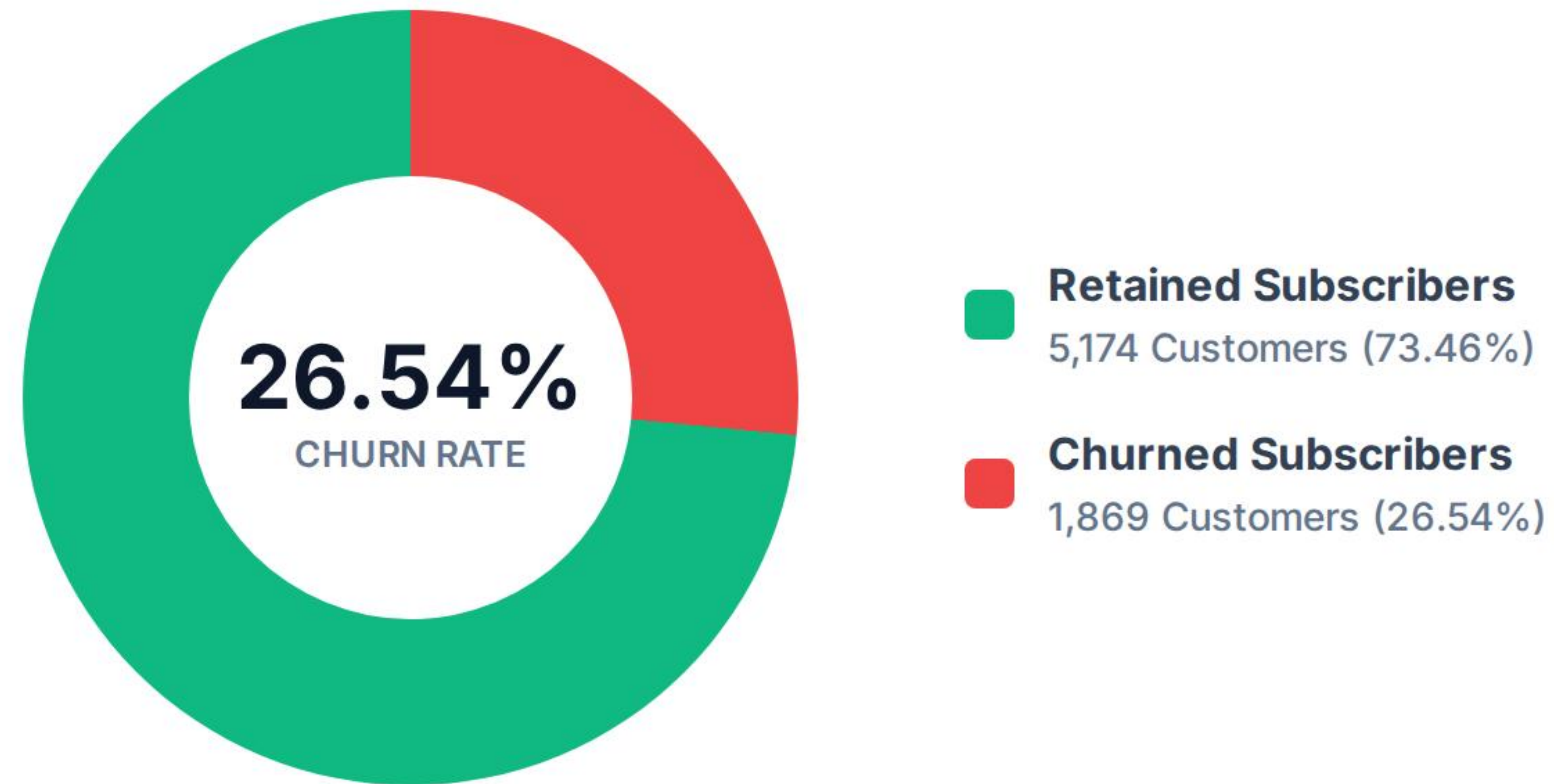
Data Cleaning & Quality Pipeline

Check / Process	Condition Identified	Transformation Applied	Impact
Missing Values	11 blank space strings in TotalCharges	Converted to numeric float; imputed 0.00 for tenure=0 accounts	100% complete numerical dataset
Duplicate Audit	0 duplicate customer records detected	Verified unique primary key constraint on customerID	No duplicate skew in analysis
Type Casting	TotalCharges parsed as string object	Cast object datatype to float64; encoded binary variables	Enabled statistical calculation
Feature Engineering	Raw continuous Tenure variable (0–72 months)	Segmented into 12-month cohort buckets for risk analysis	Revealed tenure risk curve

Executive Portfolio KPIs



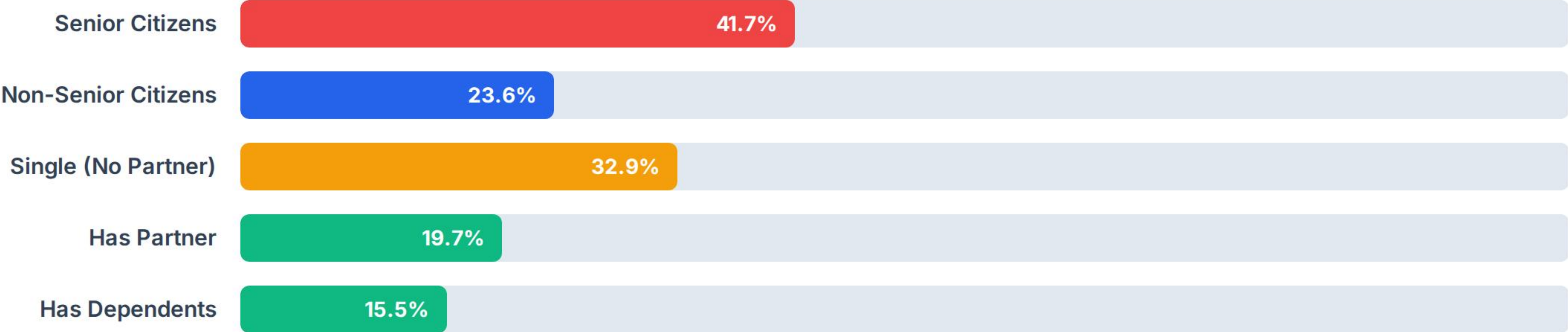
Overall Churn Distribution



💡 Key Takeaway: More than 1 out of every 4 customers churns. Addressing top risk pools is vital to stopping subscription loss.

Customer Demographics & Attrition

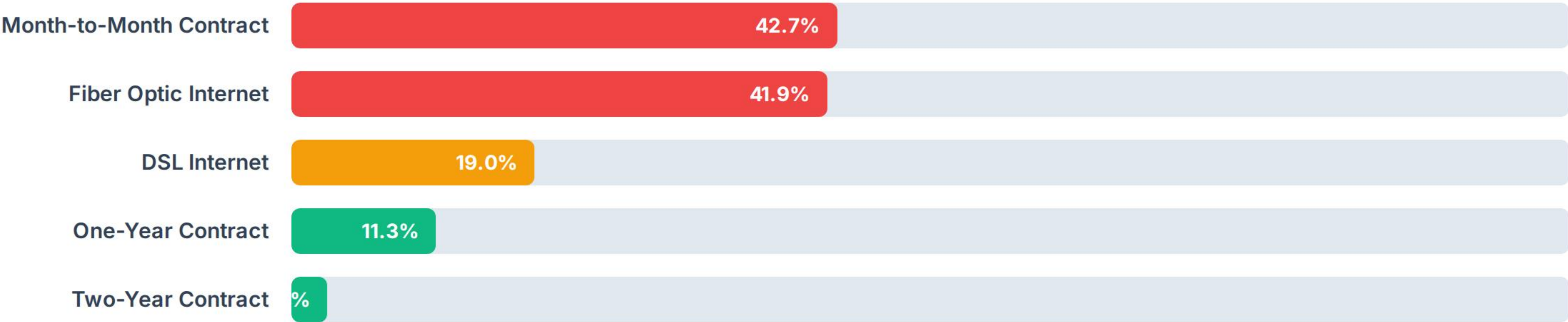
Churn Rate by Demographic Cohort



📊 Insight: Senior Citizens suffer a high churn rate of 41.7% (nearly double non-seniors). Unattached single subscribers also show higher risk.

Service Type & Contract Analysis

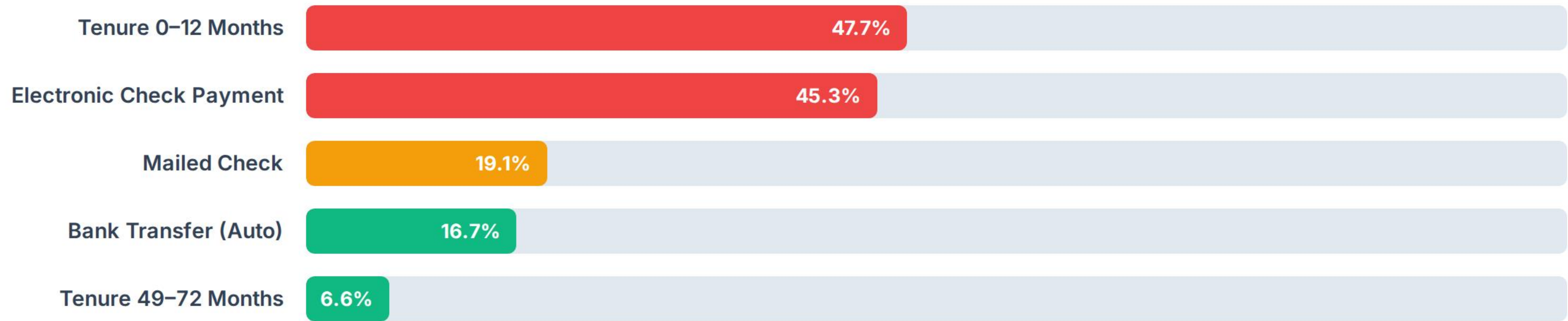
Churn Rate by Service & Contract Structure



⚠ Critical Factor: Month-to-month contracts (42.7% churn) and Fiber Optic users (41.9% churn) are the main drivers of lost revenue.

Billing Methods & Tenure Risk

Churn Rate by Payment Method & Tenure Bucket



⚡ Revenue Leakage: Electronic check users churn at 45.3% (vs ~16% for auto-pay). New accounts (<1 yr) face 47.7% churn risk.

Key Insights & Core Risk Factors

Contract & Onboarding Gaps

Short-Term Contract Volatility: 55% of total subscribers (3,875 accounts) are on Month-to-Month contracts, exhibiting a severe 42.7% churn rate.

First-Year Drop-Off: 47.7% of accounts churn in their first 12 months, pointing to early onboarding friction and unoptimized initial customer experiences.

Payment Friction & Service Pricing

Electronic Check Vulnerability: Electronic Check users (2,365 accounts) churn at 45.3%, compared to <17% for automated credit card or bank transfer users.

Fiber Optic Value Disconnect: Fiber Optic subscribers suffer a 41.9% churn rate despite paying higher average monthly fees (~\$91/mo), indicating service or value issues.

Strategic Action Plan



Incentivize Annual Contract Upgrades: Offer a \$5–\$10/month loyalty discount to transition 3,875 Month-to-Month accounts onto 1-Year or 2-Year plans, locking in churn rates below 12%.



Promote Automated Auto-Pay Migration: Implement a \$10 one-time billing credit to migrate 2,365 Electronic Check users to automated Bank Transfer / Credit Card billing.



Enhance First-Year Customer Onboarding: Deploy proactive check-ins at months 1, 3, and 6 with guided technical onboarding to reduce early 47.7% churn drop-offs.



Fiber Optic Value & Support Package: Address Fiber Optic service quality and bundle free Tech Support & Online Security to justify higher pricing and retain 3,096 users.

Executive Summary & Projected ROI

Project Summary

Scope: Evaluated 7,043 subscriber accounts and identified \$2.86M in annual revenue loss driven by 1,869 churned customers (26.54% churn rate).

Primary Root Causes: Month-to-month contract dependence, manual electronic check payments, high initial first-year drop-offs, and Fiber Optic price/value friction.

Targeted Business Impact

Goal: Reduce overall customer churn rate from **26.54% down to 15.0%** through targeted retention programs.

Revenue Protection: Protecting churned subscribers can recover up to **~\$1.4M annually** in recurring revenue while increasing Customer Lifetime Value (CLTV).

Image Sources



<https://www.ey.com/content/dam/ey-unified-site/ey-com/en-gl/insights/images/ey-ai-high-tech-telecommunications-operational-data-center-in-neon-lights.jpg>

Source: www.ey.com